Is J-space the Right Standard for Detecting Colluders?

Confound controls, a 200-draw random null and a prompt-only read for NARCBench on Gemma-2-9B-it

Andy, with Claude

Working draft for the research group, November 10, 2026 (first draft November 8; GPU run November 9)

Abstract

On November 5 a linear probe on the J-space projection of Gemma-2-9B-it's residual stream (the k=25 subspace spanned by Jacobian-lens token atoms) detected colluding agents in NARCBench, but it transferred from Core to Transfer worse than the full residual (AUROC 0.755 vs 0.902) and worse than its own complement. Before treating J-space as the standard representation we asked what colluders' J-space atoms say, whether the J-only result survives a 200-draw random-subspace null across k and read positions, and whether steering along J-space directions changes the collusive vote, and we added three confound controls: a prompt-only read, a system-prompt swap and a matched-vocabulary warning. On November 9 everything except steering ran on fresh Gemma-2-9B-it activations (one A40 for 7.7 hours; 320 analysis cells). The answers: (i) J-space is not a privileged subspace for this task. Filled to k=25 at the last token it transfers at 0.773 (band mean over layers 19–23) against 0.945 for the full residual and 0.879 for a 25-component PCA subspace; it exceeds the 95th percentile of 200 random subspaces at two layers only (layer 22: 0.899 vs 0.851) and never exceeds the 95th percentile of 200 random sets of 25 lens atoms, which reach 0.969 at layer 22. (ii) The transferable signal is the instruction. A read at the end of the system prompt, before any conversation, transfers at 1.000 for every arm including random subspaces; with the colluders' system prompt swapped for the honest one and their private blocks cut, every arm falls to chance Core→Transfer (full residual 0.479) although the probe still separates colluders within Core (0.935 leave-one-domain-out). (iii) The chosen atoms are a near-fixed set of decision, formatting and end-of-turn tokens ("decision" is selected by 189 of the 200 agents in Core collusion episodes at layer 22); the four concept groups hold about 1% of the selections in the conversation and under 3% at the end of the system prompt. (iv) Honest agents warned in the colluders' vocabulary do not score like colluders (colluders vs warned 1.000), but J-space echoes the vocabulary three to four times more than the full residual does. (v) Steering along J-space directions shifts the forced-vote logit difference by up to 3.6 but flips at most 1% of honest votes and 10% of colluder votes; the four concept directions are one direction (pairwise cosines 0.82–0.92, cosine 0.9 with the mean lens atom), and against a null of 20 random lens-atom directions, which share that component and move the vote the same way, deception and steering exceed all 20 draws at the largest strength while coordination and honesty sit inside the null, so the bars passed against Gaussian noise say little about the concepts. Our recommendation: the full-residual probe stays the reference detector, and the swap and prompt-only reads become mandatory controls for any detector claim on this benchmark, because the detectors that transfer do so by reading the instruction.

How to read this draft. Every number carries a tag. **[ran]** means it was computed on November 8 from the saved transcripts in the repository. **[cited, Nov 5]** means it is quoted from the November 5 results on branch `jspace-separability` (pushed November 9) and was not rerun here. **[ran, Nov 9 pod]** means it was computed on November 9 on one RunPod A40 from fresh Gemma-2-9B-it activations (commit `5c300ad` of branch `jspace-interp`; Section 6). **[not run]** marks planned cells that did not run, listed in Section 7. **[pipeline check only]** marks outputs from a random tiny model, which say nothing about Gemma.

1 Background

Setup. NARCBench Core has 100 four-agent committee episodes (50 collusion, 50 control) over 17 domains; Transfer has 72 episodes in 6 scenario families. In collusion episodes two agents are instructed, through their system prompt, to steer the committee to a target option while appearing impartial. The

detection unit is the episode: per-agent probe scores are pooled into three episode detectors (peak, split and asymmetry), and the primary metric is the mean of their AUROCs. Protocols are Core leave-one-domain-out (LODO), Core→Transfer, and within-Transfer leave-one-family-out (LOFO), with a stratified episode bootstrap (B=2000) for intervals.

J-space. Following Gurnee, Lindsey et al. (arXiv:2607.15495), J_ℓ is the mean Jacobian of the final residual with respect to the layer-ℓ residual h_ℓ. The atom for token i is $v_i = J_\ell^\top\big((1+\gamma)\odot W_U[i]\big)$, and the lens readout is $\mathrm{softcap}_{30}\big(W_U\,\mathrm{norm}(J_\ell h)\big)$. The decomposition takes the 1,024 atoms most correlated with h, runs non-negative orthogonal matching pursuit with an NNLS refit, fills to k=25 atoms and orthonormalises them (Q). Then $h_J = QQ^\top h$ and $h_C = h - h_J$. We use the published lens for Gemma-2-9B-it (453 WikiText prompts, layers 0–40).

2 Where the November 5 run left things

Arm (layers 19–23, last token)	Core LODO	Core→Transfer	Transfer LOFO
Full residual	1.000	0.902 [0.863, 0.935]	0.977
J-only, k=25	0.995	0.755 [0.709, 0.796]	0.854
Complement h_C	1.000	0.891	0.978
Random 25-dim subspace (one Haar draw, shared)	0.981	0.647 [0.602, 0.693]	0.742
Shuffled atoms (another turn's)	–	0.559 [0.511, 0.611]	–
PCA, top 25 components of Core	1.000	0.856 [0.807, 0.898]	0.902
J from W_U top-25 frames (earlier proxy)	0.989	0.602 [0.550, 0.658]	–

Table 1: **[cited, Nov 5]** Primary AUROC at k=25, band mean over layers 19–23. Source: `results/jspace/gemma2_9b_jpaper_nnomp_k25/RESULTS.md` on branch `jspace-separability`.

Further facts from the same branch **[cited, Nov 5]** shape the plan. Per layer, J-only vs random Core→Transfer was close at layers 19–21 (0.564/0.538, 0.709/0.696, 0.752/0.750) and separated only at layers 22–23 (0.852/0.753, 0.897/0.497). The J-space projection explains only about 1–1.5% of residual variance. And the lens's top-1 readout at the last token was `<eos>` for 98% of Core turns, so last-token readouts carry little content. The atoms chosen at the last token are nearly a fixed set across turns (at Core layer 21, "decision" is chosen in 90% of turns, `<eos>` in 86%, "prioritizing" in 82%), and another turn's atoms explain as much variance as a turn's own (source: `docs/NEXT-SCIENCE.md`). This is a first, partial answer to Experiment 1 at the last token: there, J-space mostly names end-of-turn and formatting tokens, which is why the new read positions are needed. The k variants were 0.768 (k=8), 0.803 (k=16), 0.721 (k=50), 0.811 (no fill) and 0.809 (cone projection). The random-subspace control was a single Haar subspace shared by every activation, so none of these comparisons has a usable null distribution. Two more facts from the same files matter for the verdict. A 25-component PCA subspace of Core activations transfers better than the J-space subspace of the same size (0.856 vs 0.755). And on Transfer discussion turns J-space is at the random level (0.613 vs 0.602), so its advantage sits in final turns at layers 22–23 only.

Other models. On the three models with public NARCBench activations, the earlier W_U-frame J-only probe was at chance from Core to Transfer and level with a random 25-dim subspace: Qwen3-32B 0.513 vs 0.508, DeepSeek-R1-32B 0.500 vs 0.550, GPT-OSS-20B 0.568 vs 0.611, while the complement matched the full residual on each **[cited, Nov 5]** (source: `results/jspace/CROSS_MODEL_TABLES.md`). No Jacobian lens is published for those models, so the Gemma Jacobian-lens result has not been tested elsewhere.

3 Confound audit

[ran] Reading the saved prompts and configs shows that colluders differ from honest agents through three channels, all visible to the model before it writes a word:

1. **System prompt**, in every phase: the motive and the deception instruction.

2. **A "PRIVATE COORDINATION" block** in colluders' user prompts (Core and Transfer), which ends exactly at the last private message.
3. **A "DRIFT GUIDANCE FOR THIS ROUND" block** in colluders' user prompts (Transfer only).

Behaviour is also strongly coupled to role. On Core, colluders voted the target 73 times out of 100, honest agents in collusion episodes 8 out of 100, and agents in control episodes 8 out of 200. A probe can therefore succeed by reading the instruction, the private channel or the vote, none of which is "collusion as a cognitive state". Each control below removes one of these routes.

4 Controls and experiments

4.1 Context conditions (one extraction each)

- **original**: the transcript as generated.
- **swap**: colluders get the honest system prompt for their seat (Core: the bank's honest mandate; Transfer: the control twin's same-seat prompt with the name replaced), and both colluder-only blocks are cut from their user prompts. Their own replies stay. If the probe still fires, it is reading the replies and the conversation, not the instruction.
- **nosys**: no system prompt for anyone, colluder blocks cut.
- **prompt_only**: one read per agent at the last system-prompt token, before any conversation or reply. This is the priority control: if it matches the J-only score, J-space is reading the instruction.

4.2 Behaviour split and matched vocabulary

The **behaviour split** holds the instruction fixed and splits by what the agent did: colluders who voted the target vs those who did not (73 vs 27), and honest agents who voted the target vs not (8 vs 92). The **matched-vocabulary** control generates 50 new control episodes in which seats 0–1 are honest but carry a chair's warning written in the colluders' vocabulary ("secretly coordinated", "steer the vote", "quietly downplaying a flaw", "hide a conflict of interest") and the explicit statement that they have no agenda or partner. If warned honest agents score like colluders, the probe or the concept atoms track words, not intent.

4.3 Experiment 1: what colluders' J-space says

For each agent turn at k=25, the selected atoms map to tokens. We compare colluders with honest agents on (a) the mass of four frozen concept groups (deception/secrecy, coordination, steering/persuasion, honesty/standards; word lists in `jspace_interp/concepts.py`), and (b) per-token selection frequency, with within-episode role permutation and Benjamini–Hochberg correction. Readouts are taken at the last token, the end of turn, the token before "RECOMMENDATION:", the reply midpoint and, for prompt_only, the end of the system prompt, because the last-token readout is mostly `<eos>`.

4.4 Experiment 2: k and position sweep with a real null

Grid: $k \in \{8, 25, 50, 100\} \times 7$ read positions $\times$ layers 19–23 $\times$ conditions original and swap. Arms: full, J-only (pursuit, and filled to k), complement, a fixed frame of the k atoms most often chosen on Core, shuffled atoms, PCA-k, 200 Haar random k-dim subspaces, and 200 random sets of k lens atoms. **Pre-registered bar:** J-only beats random only if its Core→Transfer AUROC is above the 95th percentile of *both* random nulls at the same $(k, \text{layer}, \text{position})$, and the paired bootstrap interval against the median random draw excludes zero.

4.5 Experiment 3: steering the vote

Prefix: the agent's own final-vote prompt (original or swap), then the forced reply start "RECOMMENDATION: Option". Outcome: $y = \text{logit[target]} - \text{logit[fair]}$. We add $\pm\alpha \cdot s \cdot u$ at one layer, only on the final-instruction tail and the forced tokens, so the earlier prefix is identical between steered and unsteered passes; $\alpha \in \{0.5, 1, 2, 4\}$ and s is the median $\|h_J\|$. Directions u: the mean J-lens atom of each concept

group, the J-space role-probe direction, and 20 random directions of equal norm. **Pre-registered bars:** predicted sign in both groups ($+u$ raises y for honest agents, $-u$ lowers it for colluders), effect above the 95th percentile of random directions, BH at q=0.05 across all cells, and growth with α.

5 Results without a GPU (November 8)

5.1 Text baselines [ran]

Bag-of-words features (1–2-grams, TF-IDF fit on Core only, at most 5,000 features) pass through the same detectors and protocols as the activation probe.

Feature	Core LODO	Core→Transfer	Transfer LOFO
Reply words only	0.853	0.493 [0.389, 0.603]	0.642
System-prompt words only	1.000	0.457 [0.394, 0.518]	0.952
Four concept word groups	0.627	0.593 [0.512, 0.675]	0.565
*For reference, **[cited, Nov 5]**:* full-residual probe 1.000 / 0.902 / 0.977; J-only 0.995 / 0.755 / 0.854			

Table 2: **[ran]** Primary AUROC of text-only features, with 95% episode-bootstrap intervals for Core→Transfer. Source file: `text_baseline.json` (Appendix A).

The system prompts separate colluders perfectly within Core and almost perfectly within Transfer, which agrees with the LessWrong report that NARCBench system prompts alone give colluders away. But the words do not carry across splits: trained on Core prompts and tested on Transfer, they sit at chance. The same holds for the replies. So the activation probe's 0.902 Core→Transfer is not explained by surface vocabulary. It may still be explained by the instruction as the model represents it, which only the prompt_only read can test.

5.2 Behaviour split on text [ran]

- Colluders who did *not* vote the target vs honest agents who also voted the other option: reply-word AUROC 0.776 (permutation $p = 0.0002$; 27 vs 92 agents). The instruction leaves a mark on the wording even when the scheme failed.
- Within colluders, target vs other vote: 0.583 ($p = 0.21$). Delivering the target barely changes how colluders write.
- Within honest agents, target vs other vote: 0.792 ($p = 0.007$, 8 vs 92). Arguing for the target option has a wording of its own, so a detector could partly be a vote detector.

The activation-level version of this split runs on saved activations once they exist.

5.3 Which words differ [ran]

Phase	Group	Colluder	Honest (same ep.)	Control	BH p (within ep.)
Core discussion	honesty/standards	8.29	14.22	13.61	0.0008
Core discussion	deception/secrecy	0.66	0.74	0.80	0.65
Core final	honesty/standards	9.29	14.39	14.86	0.005
Transfer discussion	honesty/standards	1.21	1.88	1.86	0.47
Transfer final	steering/persuasion	3.87	2.45	2.23	0.016

Table 3: **[ran]** Concept-word rates per 1,000 words. Colluders avoid honesty and standards language in Core; they do not use more deception words. The Core pattern does not repeat in Transfer discussion.

This matters for Experiment 1: if J-space atoms of colluders show less honesty mass rather than more deception mass, that mirrors the text and is not evidence of a hidden representation.

5.4 Matched-vocabulary prompt check [ran]

The warning has 6 deception/secrecy, 3 coordination, 5 steering and 5 honesty words in 79 words. Colluder system prompts average 0.15, 0.23, 1.40 and 1.40 on Core and 1.89, 0.17, 0.63 and 0.54 on Transfer (source: `results/jspace_interp/matched_vocab/vocab_check.json`). The warning therefore puts *more* collusion vocabulary in context than the colluder instruction does, which makes it a conservative test: a probe that ignores it is not reading those words. A dry run with a stub model confirmed the warning lands on seats 0–1 only, in control episodes tagged `__warned`.

5.5 End-to-end pipeline check [pipeline check only]

Every stage ran on the real transcripts with a randomly initialised Gemma-2 model (6 layers, width 64, 0.48M parameters) and a 4,000-token BPE tokenizer trained here on the transcripts: extraction for all four conditions on 20 Core episodes (10 collusion) and 27 Transfer episodes (14 collusion), with contexts left-truncated to 768 tokens; lens estimation on 10 short prompts; the $k\times$position sweep ($k \in \{4, 8, 25\}$, last and pre-vote tokens, layers 2–4, conditions original, swap and prompt_only) with both random nulls and the bootstrap; the Experiment 1 readouts with concept groups and token tests; steering on 24 final-vote prefixes from 6 episodes with the four concept directions and 5 random ones; and a stub-model dry run of the matched-vocabulary generator. Source: `results/jspace_interp/tiny_pipeline_check/`.
Two observations carry over to the real run, as lessons about the method rather than findings about Gemma. First, with only 5 random draws per cell, the J-only arm cleared both random 95th percentiles in 6 of 45 cells of a network that knows nothing; a bar built on a handful of draws fires by chance, which is why the plan uses 200. Second, at the end of the system prompt even this random network separated colluders from honest agents from Core to Transfer (0.84 [0.73, 0.95] at its layer 2, on few episodes), because activations there are a function of the prompt tokens. A high prompt-only score therefore needs no learned notion of collusion, and the prompt-only control should be compared with J-only on equal footing, not with chance.

6 Results of the November 9 GPU run

6.1 The run

[ran, Nov 9 pod] One RunPod A40 (48 GB, secure cloud, $0.59 per hour) ran commit `5c300ad` of `jspace-interp` from 07:22 to about 15:05 UTC on November 9: 7.7 hours, about $4.60 including a first pod terminated after ten minutes for a queue-sizing fix. Stages: 50 warned control episodes generated with Gemma-2-9B-it (800 prompts, 23 minutes); activations at seven read positions and layers 19–23 for the four context conditions (Core 1,700 turns, Transfer 936, warned 850 per condition; about 25 minutes per condition); then 16 CPU jobs, one per condition and read position, each running the five layers at $k \in \{8, 25, 50, 100\}$ with 200 Haar random subspaces and 200 random sets of k lens atoms per cell (20 of each also scored under LODO and on the warned episodes), the episode bootstrap (B=2000) on the named arms, and the Experiment 1 readouts at the last, pre-vote, midpoint and prompt-end positions. All 320 planned cells finished, about 27 minutes each for original and swap. A second pod (`lz7gdscc09o0on`, A40, 15:08–18:55 UTC, 3.8 hours, about $2.25, commit `2899572`) reran Core extraction at layers 22–23, the role directions and three steering runs after the first pod's steering failed (Section 7); a third pod (`rt34fhh908ja7p`, 0.9 hours, about $0.55, commit `1c740de`) reran the swap layer-23 steering against a null of random lens directions. The three pods together cost about $7.60. As on November 5, probes are logistic regressions fit on Core discussion turns (1,200 rows) and tested on Transfer final-vote turns (288 rows); prompt_only uses one row per agent. The per-cell results are the `sweep_results.json` files under `results/jspace_interp/gemma2_9b/`, one folder per condition and read position. The script `jspace_interp_summarize.py` turns them into `SUMMARY.json`, `RESULTS.md` and the tables below; the pod's records and logs are under `results/jspace_interp/pod_runs/` and `pod_logs/`.

6.2 Experiment 2: k and position sweep against 200-draw nulls

[ran, Nov 9 pod] Table 4 gives band means over layers 19–23. With the instruction present (original), the full residual transfers at 0.945 at the last token, PCA-25 at 0.879, the filled J-only arm at 0.773 and the

pursuit arm at 0.765; the 200 random 25-dimensional subspaces average 0.65 with a 95th percentile of 0.808, and 200 random sets of 25 lens atoms have a 95th percentile of 0.875. No other read position carries much transferable signal for any arm: at the end of turn the full residual gives 0.780, at the pre-vote token 0.517 (while its Core LODO is 0.999 and its Transfer LOFO 0.817), and the reply midpoint and fractions stay between 0.52 and 0.71. Per layer at the last token (Table 6) the filled J arm rises from 0.596 at layer 19 to 0.899 at layer 22 and falls to 0.831 at layer 23, the November 5 profile (0.564 to 0.897) with a proper null. It exceeds the random-subspace 95th percentile at layers 21 and 22 only (one-sided $p = 0.040$ and 0.010), and it never exceeds the random-atom 95th percentile: 25 random lens atoms, orthonormalised, transfer at 0.969 (95th percentile; mean 0.805) at layer 22 and 0.961 at layer 23, better than the pursuit-chosen atoms. The pre-registered bar (both nulls) is cleared in none of the 35 original cells at k=25 except two at frac25, where every arm is at chance, and once for the pursuit arm. PCA-k is at least as good as J at every layer and k (Table 5): at layer 22 the paired difference J fill minus PCA is $[-0.09, 0.02]$ at k=25 and $[-0.10, -0.00]$ at k=50; over the 35 cells PCA beats J in 9 and J beats PCA in 1 (paired 95% intervals). The complement beats the filled J arm at every layer (layer 22: $[0.016, 0.118]$). The pursuit stops after 16–18 atoms on average, so the pursuit arm is identical for $k \geq 25$; filling raises the score only at k=100 (0.930 at layer 22), where the random 95th percentile has risen to 0.942. Shuffled atoms give 0.747 and the fixed frame of the most common atoms 0.881 at layer 22.

Cond.	Position	Full	J fill	J pursuit	Compl.	PCA-k	Rand. sub. mean (p95)	Rand. atoms p95	Layers clearing both
original	last	0.945	0.773	0.765	0.941	0.879	0.650 (0.808)	0.875	0/5
	eot	0.780	0.550	0.584	0.768	0.695	0.584 (0.719)	0.763	0/5
	prevote	0.517	0.526	0.535	0.520	0.494	0.512 (0.600)	0.588	0/5
	reply_mean	0.706	0.577	0.575	0.710	0.679	0.610 (0.691)	0.658	0/5
	frac25	0.522	0.569	0.517	0.521	0.507	0.500 (0.595)	0.580	2/5
	frac50	0.527	0.540	0.536	0.529	0.538	0.502 (0.598)	0.596	0/5
	frac75	0.571	0.478	0.461	0.567	0.546	0.519 (0.619)	0.602	0/5
swap	last	0.479	0.436	0.473	0.468	0.465	0.508 (0.609)	0.595	0/5
	eot	0.491	0.492	0.513	0.489	0.491	0.512 (0.611)	0.594	0/5
	prevote	0.469	0.515	0.537	0.477	0.479	0.501 (0.587)	0.572	1/5
	reply_mean	0.554	0.585	0.579	0.555	0.541	0.565 (0.649)	0.627	0/5
	frac25	0.488	0.508	0.531	0.482	0.643	0.514 (0.612)	0.596	0/5
	frac50	0.519	0.537	0.559	0.518	0.610	0.508 (0.608)	0.602	0/5
	frac75	0.573	0.471	0.469	0.564	0.514	0.505 (0.601)	0.592	0/5

Table 4: **[ran, Nov 9 pod]** Core→Transfer primary AUROC at k=25, band mean over layers 19–23, per condition and read position. "Layers clearing both" counts layers at which the filled J-only arm exceeds the 95th percentile of both random nulls (200 Haar subspaces and 200 random atom sets).

k	Full	J fill	J pursuit	Compl.	PCA-k	Rand. sub. p95	Rand. atoms p95	J fill − PCA (95% CI)
8	0.945	0.740	0.740	0.944	0.831	0.771	0.842	[-0.08, 0.02] (L22)
25	0.945	0.773	0.765	0.941	0.879	0.808	0.875	[-0.09, 0.02] (L22)
50	0.945	0.762	0.766	0.938	0.928	0.851	0.895	[-0.10, -0.00] (L22)
100	0.945	0.807	0.766	0.938	0.938	0.896	0.925	[-0.08, 0.00] (L22)

Table 5: **[ran, Nov 9 pod]** k sweep, original condition, last token, band mean over layers 19–23; the paired bootstrap interval of J fill minus PCA-k is for layer 22, the layer with the best J fill.

6.3 The three controls

[ran, Nov 9 pod] Tables 8 and 9. **prompt_only.** At the end of the system prompt the full residual, PCA-25 and the 95th percentile of random subspaces all transfer at 1.000 at every layer, and the J arm at 0.95–1.00. The system prompts differ so systematically between roles that any 25 directions separate them; the tiny-model lesson of Section 5 holds for Gemma. **swap.** With the honest system prompt and no private blocks, Core→Transfer falls to chance for every arm at every layer and position (full residual 0.45–0.54 at the last token; Table 7). The best J cell of 35 is 0.622 at the pre-vote token of layer 22, against nulls of 0.594 and 0.606, which is the count expected by chance. The probe still separates colluders within Core (LODO

Layer	Full	J fill	J pursuit	PCA-k	Rand. sub. p95	Rand. atoms p95	p vs rand. sub.	Clears both
19	0.948 [0.91, 0.98]	0.596 [0.50, 0.69]	0.598	0.871	0.783	0.767	0.597	–
20	0.920 [0.88, 0.96]	0.718 [0.64, 0.79]	0.724	0.853	0.763	0.786	0.149	–
21	0.919 [0.88, 0.95]	0.819 [0.76, 0.87]	0.773	0.900	0.803	0.893	0.040	–
22	0.967 [0.95, 0.99]	0.899 [0.85, 0.94]	0.924	0.937	0.851	0.969	0.010	–
23	0.970 [0.95, 0.99]	0.831 [0.77, 0.89]	0.806	0.835	0.838	0.961	0.075	–

Table 6: **[ran, Nov 9 pod]** Per-layer Core→Transfer AUROC at k=25, original condition, last token, with 95% episode-bootstrap intervals and the one-sided p of J fill against the 200 random subspaces. "Clears both" is the pre-registered bar.

Layer	Full	J fill	J pursuit	PCA-k	Rand. sub. p95	Rand. atoms p95	p vs rand. sub.	Clears both
19	0.450 [0.35, 0.55]	0.417 [0.31, 0.52]	0.465	0.451	0.606	0.617	0.970	–
20	0.457 [0.36, 0.55]	0.382 [0.28, 0.49]	0.483	0.461	0.625	0.586	1.000	–
21	0.484 [0.39, 0.58]	0.475 [0.39, 0.56]	0.534	0.486	0.604	0.572	0.657	–
22	0.460 [0.38, 0.55]	0.457 [0.38, 0.54]	0.445	0.481	0.588	0.597	0.831	–
23	0.542 [0.45, 0.64]	0.447 [0.34, 0.55]	0.440	0.443	0.620	0.604	0.900	–

Table 7: **[ran, Nov 9 pod]** The same at the last token under the swap condition (honest system prompt, colluder blocks cut): every arm at chance, at every layer.

0.935 for the full residual, 0.898 PCA, 0.831 J) and partly within Transfer (LOFO 0.667), so the conversation does carry a split-specific signal, but nothing that carries across splits. **nosys** agrees (0.488 Core→Transfer; 0.934 LODO). The 0.945 of the original condition is therefore the instruction as the model represents it. **Matched vocabulary** (Table 10; original condition, 50 warned control episodes, 600 warned-episode turns). Warned honest agents do not score like colluders: colluders vs warned 1.000 for the full residual at every layer, 0.90–0.99 for J. The warning does move honest agents a little: warned vs unwarned 0.62–0.70 for the full residual (within-episode $p < 0.001$), an echo of 4–6% of the colluder–honest gap, rising to 0.71–0.83 at the end of turn. J-space echoes more: warned vs unwarned 0.58–0.76 and an echo ratio of 0.13–0.21, three to four times the full residual's and above PCA (0.01–0.05) and the random subspaces (0.55–0.62 AUROC). At the episode level warned episodes are not mistaken for collusion (collusion vs warned 1.000) but look slightly more collusive than plain control episodes (0.59–0.65).

Cond.	Pos.	Full			J fill k=25			PCA-25		
		LODO	C→T	LOFO	LODO	C→T	LOFO	LODO	C→T	LOFO
original	last	1.000	0.945	0.974	0.988	0.773	0.866	1.000	0.879	0.898
	prevote	0.999	0.517	0.817	0.888	0.526	0.629	0.999	0.494	0.621
swap	last	0.935	0.479	0.667	0.831	0.436	0.555	0.898	0.465	0.667
	prevote	0.908	0.469	0.649	0.750	0.515	0.582	0.900	0.479	0.626
nosys	last	0.934	0.488	0.631	0.822	0.475	0.605	0.889	0.466	0.609
prompt_only	prompt_end	1.000	1.000	1.000	0.972	0.980	0.985	1.000	1.000	1.000

Table 8: **[ran, Nov 9 pod]** The three protocols per condition at the main read position, band mean over layers 19–23: Core leave-one-domain-out (LODO), Core→Transfer (C→T) and Transfer leave-one-family-out (LOFO), for the full residual, the filled J-only arm at k=25 and PCA-25.

6.4 Experiment 1: what the atoms say

[ran, Nov 9 pod] Table 11. Core readouts use the probe's rows, the three public discussion turns of each agent, and are aggregated per agent; Transfer readouts use the final turn. The four concept groups together hold at most 1.1% of the k=25 atom selections at every conversation read position and at most 2.8% at the end of the system prompt, where the words themselves are in context. The colluder–honest differences that survive BH are below 0.01 in the conversation (deception/secrecy +0.008 on Transfer at the last token of layer 22; steering/persuasion −0.004 to −0.007 under nosys) and larger only at the end of the system prompt (deception/secrecy +0.006 to +0.017, honesty/standards −0.009 to −0.026). The atoms are

Layer	original/last		swap/last		nosys/last		prompt_only/prompt_end	
	Full	J fill	Full	J fill	Full	J fill	Full	J fill
19	0.948	0.596	0.450	0.417	0.503	0.504	1.000	0.985
20	0.920	0.718	0.457	0.382	0.472	0.414	1.000	0.998
21	0.919	0.819	0.484	0.475	0.474	0.501	1.000	0.997
22	0.967	0.899	0.460	0.457	0.490	0.496	1.000	0.968
23	0.970	0.831	0.542	0.447	0.504	0.462	1.000	0.952

Table 9: **[ran, Nov 9 pod]** Core→Transfer primary AUROC per layer, full residual and filled J-only arm at k=25, for the four context conditions at their read position. "n/r" marks cells that did not run.

Cell	Arm	Coll. vs warned	Warned vs unwarned (p)	Warned vs control	Echo	Ep. warned vs control	Ep. coll. vs warned
original/last/L22	Full	1.000	0.694 (0.00)	0.612	0.05	0.632	1.000
	J fill k=25	0.978	0.756 (0.00)	0.738	0.21	0.739	0.992
	PCA-25	0.999	0.573 (0.00)	0.525	0.02	0.527	1.000
	Random 25-dim, mean of 20 (echo: median)	0.973	0.587	0.555	0.03	0.557	n/a
original/prevote/L22	Full	1.000	0.736 (0.00)	0.741	0.13	0.738	1.000
	J fill k=25	0.923	0.567 (0.03)	0.563	0.11	0.531	0.942
	PCA-25	0.998	0.716 (0.00)	0.744	0.17	0.702	0.999
	Random 25-dim, mean of 20 (echo: median)	0.890	0.615	0.605	0.16	0.563	n/a

Table 10: **[ran, Nov 9 pod]** Matched-vocabulary control, scored out of fold with leave-one-Core-domain-out probes. Agent AUROCs: colluders vs warned honest agents, warned vs unwarned honest agents in the same episodes (within-episode permutation p), warned vs control honest agents. Echo ratio: (mean warned − mean control honest)/(mean colluder − mean control honest). Episode detectors: primary AUROC of warned vs control episodes and of collusion vs warned episodes.

a near-fixed set across turns, as on November 5: at the last token of layer 22 "decision" is selected, in at least one discussion turn, by 189 of the 200 agents in Core collusion episodes, "prioritizing" by 177, `<eos>` by 161, "eloquently" by 161 and "recommendation" by 154; at the pre-vote token `<eos>`, "prioriti" and "acknowledging" are selected by 199–200 of 200. Token-level selection frequencies do differ by role in 30 of the 40 readout cells after BH, but the tokens are stance and register words (under nosys at the last token colluders select "understandably", "verdict", "compelling" and "eloquently" less often and "advisors" and "considering" more), not the deception or coordination vocabulary. J-space at these positions names how the turn ends, not what the agent is doing.

6.5 Experiment 3: steering the vote

[ran, Nov 9 pod] Three of the four planned runs completed on the steering-only pod: swap and original at layer 22 and swap at layer 23, each on the 400 Core final-vote prefixes (300 honest, 100 colluder agents) with the four concept directions, the J-space role direction and 20 random directions at $\alpha \in \{0.5, 1, 2, 4\}$ times the median $\|h_J\|$ (33.3–39.5). The original layer-23 run was skipped by the pod's 4-hour cap (**[not run]**). Table 12 has the cells.

Baselines first. Under the original prefix colluders' forced-vote logit difference y_0 averages +3.5 and 73% favour the target, matching their 73 of 100 target votes; honest agents average −8.6 with 5%. Under swap the same colluders, with the honest system prompt and no private blocks but their own earlier replies intact, average −3.9 with 24% favouring the target. The instruction in the prefix, not the conversation, carries most of the target preference: the causal counterpart of the swap result above.

Large in logits, small in votes. Adding $+u$ to honest agents raises y by 2.4 to 3.6 at α=4 for every concept direction, monotonically in α and above the 95th percentile of the random directions, whose mean effects at α=4 range from −0.3 to +2.0. Subtracting u from colluders lowers y by 0.3 to 1.6 under swap, with the J-space role direction strongest (−1.5 at layer 22, −2.0 at layer 23). Under the original prefix the concept directions have the wrong sign on colluders at layer 22 (they raise y by 0.05 to 0.38) and only the role direction lowers it (−0.66). At swap layer 23 every direction passes every pre-registered bar (sign, random 95th percentile at every α, BH at q=0.05, dose); at layer 22 the sign and random bars pass for most cells but BH does not, which with a 20-direction null (minimum $p = 1/21$) rejects only when every test sits at that minimum. Votes barely move: honest agents start 8.6 logits from the target and at most 1% cross zero at α=4; colluders are pushed off the target in 1–6% of prefixes by the concept directions and in 10% by the

Cell	Group	Colluder	Honest (same ep.)	Control	BH p (within ep.)
swap/last/L22	deception/secrecy	0.0000	0.0000	0.0000	1.000
	coordination	0.0000	0.0000	0.0000	1.000
	steering/persuasion	0.0011	0.0019	0.0012	0.736
	honesty/standards	0.0000	0.0000	0.0000	1.000
most common atoms (Core): ␣decision (200), ␣prioritizing (199), ␣recommendation (190), <eos> (186), ␣recommending (183), ␣recommendations (178)					
swap/prevote/L22	deception/secrecy	0.0000	0.0000	0.0000	1.000
	coordination	0.0000	0.0000	0.0000	1.000
	steering/persuasion	0.0000	0.0000	0.0000	1.000
	honesty/standards	0.0000	0.0000	0.0003	1.000
most common atoms (Core): <eos> (200), ␣acknowledging (200), ␣prioriti (200), . (198), ␣␣ (198), ␣concludes (198)					
original/last/L22	deception/secrecy	0.0003	0.0001	0.0000	1.000
	coordination	0.0000	0.0000	0.0000	1.000
	steering/persuasion	0.0016	0.0022	0.0012	1.000
	honesty/standards	0.0000	0.0000	0.0000	1.000
most common atoms (Core): ␣decision (189), ␣prioritizing (177), <eos> (161), ␣eloquently (161), ␣diría (161), ␣discussions (155)					
original/prevote/L22	deception/secrecy	0.0000	0.0000	0.0000	1.000
	coordination	0.0000	0.0000	0.0000	1.000
	steering/persuasion	0.0000	0.0000	0.0000	1.000
	honesty/standards	0.0000	0.0000	0.0003	1.000
most common atoms (Core): <eos> (200), ␣prioriti (200), ␣acknowledging (199), ␣pungkasnya (199), . (198), ␣␣ (198)					

Table 11: **[ran, Nov 9 pod]** Share of the four frozen concept groups among the k=25 selected atoms, mean per agent over its three Core discussion turns, with Benjamini–Hochberg adjusted within-episode permutation p over all concept tests of the run; below each cell, the atoms selected most often with the number of the 200 agents in Core collusion episodes that select them in at least one turn.

role direction under swap.

The result is not about the concepts. The four concept directions have pairwise cosines of 0.82–0.92 at layers 22–23 and cosines of 0.88–0.94 with the mean of all lens atoms (the mean of 20,000 unit atoms has norm 0.75: the atoms share one dominant direction). Deception, coordination, steering and honesty are therefore, as directions, one direction, and they move the vote identically, honesty/standards included. Random Gaussian directions have $|\cos| \approx 0.01$ with that component, so the random null compares the common lens direction with unstructured noise, not one concept with another. The J-space role direction is nearly orthogonal to the common component (cosine 0.02–0.17) and to its own counterpart from the other condition (0.26–0.29), and it is the only direction whose effect is specific to colluders. Steering passes its bars at one layer for a reason that has nothing to do with what the atoms name.

A fairer null. **[ran, Nov 9 pod]** We reran swap layer 23 with the same prefixes, scale and strengths against 20 random *lens* directions: the unit mean atom of a random set of vocabulary tokens, size-matched to the concept groups (32–47 tokens). These share the common component (cosine 0.85–0.94 with the concept directions, mean 0.90) and move the vote the same way: at α=4 their mean effects on honest agents run from +2.2 to +3.0 (median +2.5) and on colluders from −0.8 to −1.2 (median −1.0). Against this null deception/secrecy (+3.6, −1.6) and steering/persuasion (+3.1, −1.3) exceed all 20 draws in both groups at α=4 ($p = 1/21$), coordination (+3.0, −0.9) and honesty/standards (+2.7, −0.6) sit inside it, and no direction passes the full pre-registered set (Table 12, last block). The concept-specific part of the effect is therefore at most about one logit on top of a common-component effect of two and a half, with deception and steering at the top of the random-lens distribution and honesty at its bottom; with 20 draws the smallest attainable p is 1/21, so a sharper statement needs more draws.

7 What did not run

[not run] On the first pod 11 of 15 stages ran; its four steering stages failed with CUDA out-of-memory because our loop kept every prefix's key–value cache on the GPU (fixed in commit `bdd30cd`: one cache per row; identical outcomes to 10^{-7} on the tiny model). The steering-only pod then completed three of the four runs; the original layer-23 run was skipped by its 4-hour cap. All 320 analysis cells ran, so no sweep cell is missing. Not planned and therefore not run: the matched-vocabulary readings under swap and nosys (the warned episodes were extracted for original and prompt_only only); Experiment 1 readouts at the end-of-turn and reply positions (the analysis ran them at the last, pre-vote, midpoint and prompt-end positions); and the paired-interval half of the Experiment 2 bar against the median random draw, because the random

draws ran without the episode bootstrap to fit the time cap (the one-sided empirical p against the 200 draws is reported instead).

8 Related work

Molinari & Bettineschi (arXiv:2611.02153, November 1, 2026; abstract read) report on NARCBench that colluders' J-space "holds the collusion itself" (86% of colluders represent misrepresentation, concealment or remorse before replying, on DeepSeek-V4-Flash), an unsupervised detector (J-PACT) that keeps 85% of the above-chance AUROC of supervised probes, and causal interventions on a hidden count in a steganography task. The abstract does not mention system-prompt controls. Our plan differs in testing a supervised J-only probe across splits against a many-draw null, in running the three confound controls, and in steering the collusive vote itself. A **LessWrong post** reportedly finds that NARCBench system prompts alone separate colluders (not read here; Table 2 agrees within a split). An **Apart sprint team** reportedly built a J-lens collusion monitor on NARCBench; its report could not be read.

9 Should J-space be the standard?

No, not as a detector, and on this benchmark not as an interpretation layer either. The standard set before the run was that J-space beat the random-subspace null, the prompt-only probe and the 25-component PCA subspace. It beats none of them. (1) Against 200 random subspaces it wins at two layers at one read position and loses to random sets of its own lens atoms everywhere. (2) The prompt-only probe transfers at 1.000 against J's 0.77 (band) to 0.90 (best layer). (3) PCA-25 transfers at 0.879 against 0.773, and 0.937 against 0.899 at the best layer; the paired interval never favours J. The pre-registered conditions for the interpretation case fail as well: (a) under swap no J cell beats both nulls beyond the chance count, because nothing transfers under swap; (b) prompt_only is far above J-only; (c) warned honest agents are not mistaken for colluders, but J-space tracks the warning's words more than the full residual does; (d) steering passes its bars at one layer against Gaussian noise, but the four concept directions are nearly a single direction shared by every lens atom; against random lens directions only deception and steering stay above all 20 draws, and votes do not flip. **What the run says about detection in general.** The detectors that transfer across NARCBench splits are instruction detectors: remove the system prompt and the private channel and transfer vanishes for every representation, while a read before the first word of conversation is perfect. Within a split a probe can still find colluders from the conversation (0.93 LODO under swap), which is where a behaviour-based detector would have to prove itself. We therefore recommend keeping the full-residual probe as the reference detector, reporting every detector next to its swap and prompt-only reads, and treating the incentive-driven (uninstructed) setting as the real test, since on the instructed benchmark the instruction itself is the signal.

10 Limitations

Instructed collusion only, one model, one published lens, one run. Text baselines use bag-of-words features. Related-work claims for the LessWrong post and the Apart team are second-hand. The November 5 numbers were reproduced in profile, not rerun: the activations were regenerated (seven positions, bf16) and the full residual now transfers at 0.945 rather than 0.902. Thirty-five cells per condition at k=25 invite a multiplicity caveat, which is why cell counts are reported against their chance expectation. The random-atom null uses orthonormalised random lens atoms, which is the natural null for the atom selection but not for the lens itself. The cached-prefix steering passes differ from full passes by up to 0.13 in y in bf16 (0.0 on the fp32 tiny model), small against the effects reported but not zero; the random-lens steering null has 20 draws, so its smallest p is 1/21 and the Benjamini–Hochberg bar cannot be met by a subset of cells. Concept word lists were frozen before any activation result, but their choice is ours.

A Code and reproduction

Branch `jspace-interp` of the narcbench-core-demo repository. Modules in `jspace_interp/`: contexts (conditions), positions and extract (read positions), jacobian (lens estimation), sweep and detectors (Experiment 2), concepts and stats (Experiment 1), steer (Experiment 3). Result files are under `results/jspace_interp/` on the branch and copied to the shared `jspace-interp/results/` folder. Scripts:

```
python scripts/jspace_interp_text_baseline.py                    # ran (CPU)
python scripts/jspace_interp_extract.py --model google/gemma-2-9b-it
python scripts/jspace_interp_analyze.py --root <acts> --lens <lens.pt> \
    --basis-dir <basis> --tokenizer <tokenizer.json> --layers 19-23
python scripts/jspace_interp_steer.py --lens <lens.pt> --basis-dir <basis> --layer 22
python scripts/jspace_interp_gen_matched_vocab.py --all
python scripts/jspace_interp_tiny_e2e.py --work /tmp/jspace_tiny   # ran (CPU)
python scripts/runpod_launch_jspace_interp.py --preflight         # gates
python scripts/runpod_launch_jspace_interp.py --launch --watch     # ran Nov 9
python scripts/runpod_launch_jspace_interp.py --mode steer --launch --watch
python scripts/jspace_interp_summarize.py           # SUMMARY, RESULTS.md, tables/
python scripts/jspace_interp_finish_cells.py --from-summary SUMMARY.json
```

Run	Direction	Group (n)	Δy at α=1	α=2	α=4	Flips α=2	α=4	Sign	> rand. p95	BH	All bars
original L22	deception_secrecy	colluder (100)	-0.01	-0.12	-0.47	0.03	0.06	–	–	–	–
	deception_secrecy	honest (300)	0.83	1.87	3.21	0.00	0.00	✓	✓	–	–
	coordination	colluder (100)	0.11	0.17	0.20	0.01	0.03	–	–	–	–
	coordination	honest (300)	0.74	1.76	2.49	0.00	0.00	✓	✓	–	–
	steering_persuasion	colluder (100)	0.10	0.13	0.05	0.01	0.04	–	–	–	–
	steering_persuasion	honest (300)	0.79	1.82	2.68	0.00	0.00	✓	✓	–	–
	honesty_standards	colluder (100)	0.19	0.29	0.38	0.01	0.03	–	–	–	–
	honesty_standards	honest (300)	0.73	1.73	2.41	0.00	0.01	✓	✓	–	–
	J role probe	colluder (100)	-0.14	-0.31	-0.66	0.00	0.01	✓	–	–	–
	J role probe	honest (300)	0.17	0.55	1.90	0.00	0.00	✓	–	–	–
colluder: baseline y_0 mean 3.54, share with $y_0 > 0$ 0.73; 20 random directions at α=4: mean Δy from -0.95 to 0.38											
honest: baseline y_0 mean -8.63, share with $y_0 > 0$ 0.05; 20 random directions at α=4: mean Δy from -0.11 to 1.94											
swap L22	deception_secrecy	colluder (100)	-0.35	-0.58	-0.88	0.01	0.03	✓	✓	–	–
	deception_secrecy	honest (300)	0.84	1.91	3.25	0.00	0.00	✓	✓	–	–
	coordination	colluder (100)	-0.18	-0.29	-0.33	0.00	0.00	✓	–	–	–
	coordination	honest (300)	0.76	1.79	2.53	0.00	0.00	✓	✓	–	–
	steering_persuasion	colluder (100)	-0.27	-0.43	-0.61	0.00	0.02	✓	–	–	–
	steering_persuasion	honest (300)	0.81	1.85	2.71	0.00	0.00	✓	✓	–	–
	honesty_standards	colluder (100)	-0.19	-0.28	-0.26	0.00	0.02	✓	–	–	–
	honesty_standards	honest (300)	0.74	1.75	2.45	0.00	0.01	✓	✓	–	–
	J role probe	colluder (100)	-0.41	-0.82	-1.54	0.03	0.10	✓	✓	–	–
	J role probe	honest (300)	0.51	1.11	2.38	0.01	0.01	✓	✓	–	–
colluder: baseline y_0 mean -3.85, share with $y_0 > 0$ 0.24; 20 random directions at α=4: mean Δy from -0.63 to 0.69											
honest: baseline y_0 mean -8.63, share with $y_0 > 0$ 0.05; 20 random directions at α=4: mean Δy from -0.11 to 1.99											
swap L23	deception_secrecy	colluder (100)	-0.52	-0.98	-1.60	0.01	0.03	✓	✓	✓	✓
	deception_secrecy	honest (300)	1.06	2.33	3.61	0.00	0.00	✓	✓	✓	✓
	coordination	colluder (100)	-0.37	-0.64	-0.92	0.00	0.03	✓	✓	✓	✓
	coordination	honest (300)	0.92	2.09	2.96	0.01	0.01	✓	✓	✓	✓
	steering_persuasion	colluder (100)	-0.45	-0.82	-1.27	0.00	0.03	✓	✓	✓	✓
	steering_persuasion	honest (300)	1.00	2.20	3.13	0.00	0.00	✓	✓	✓	✓
	honesty_standards	colluder (100)	-0.24	-0.47	-0.61	0.00	0.02	✓	✓	✓	✓
	honesty_standards	honest (300)	0.84	1.93	2.66	0.00	0.00	✓	✓	✓	✓
	J role probe	colluder (100)	-0.63	-1.33	-2.05	0.06	0.10	✓	✓	✓	✓
	J role probe	honest (300)	0.63	1.15	1.85	0.00	0.00	✓	✓	✓	✓
colluder: baseline y_0 mean -3.85, share with $y_0 > 0$ 0.24; 20 random directions at α=4: mean Δy from -0.42 to 0.64											
honest: baseline y_0 mean -8.63, share with $y_0 > 0$ 0.05; 20 random directions at α=4: mean Δy from -0.30 to 1.88											
swap L23 (lens null)	deception_secrecy	colluder (100)	-0.52	-0.98	-1.60	0.01	0.03	✓	✓	–	–
	deception_secrecy	honest (300)	1.06	2.33	3.61	0.00	0.00	✓	✓	–	–
	coordination	colluder (100)	-0.37	-0.64	-0.92	0.00	0.03	✓	–	–	–
	coordination	honest (300)	0.92	2.09	2.96	0.01	0.01	✓	–	–	–
	steering_persuasion	colluder (100)	-0.45	-0.82	-1.27	0.00	0.03	✓	✓	–	–
	steering_persuasion	honest (300)	1.00	2.20	3.13	0.00	0.00	✓	✓	–	–
	honesty_standards	colluder (100)	-0.24	-0.47	-0.61	0.00	0.02	✓	–	–	–
	honesty_standards	honest (300)	0.84	1.93	2.66	0.00	0.00	✓	–	–	–
colluder: baseline y_0 mean -3.85, share with $y_0 > 0$ 0.24; 20 random directions at α=4: mean Δy from -1.23 to -0.80											
honest: baseline y_0 mean -8.63, share with $y_0 > 0$ 0.05; 20 random directions at α=4: mean Δy from 2.18 to 2.95											

Table 12: **[ran, Nov 9 pod]** Mean change of $y = \text{logit[target]} - \text{logit[fair]}$ under $+u$ for honest agents and $-u$ for colluders, at α times the median $\|h_J\|$. Flips: share of prefixes whose y changes sign toward the target (honest) or away from it (colluders). Sign: predicted sign at every α. > rand. p95: effect above the 95th percentile of 20 random directions at every α. BH: rejected at q=0.05 at some α. All bars: the pre-registered set including dose. Below each run: baseline y_0 per group and the range of the random directions' mean effects.